National GHG Inventory – Methodology Reference

Source Category Methodologies: Narrative, Coding Roadmap, and Documentation

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Introduction

This document serves three purposes:

1. **Methodology narrative** – a plain-language description of the calculation methodology for each source category, suitable for review by domain experts and auditors.
2. **Coding roadmap** – technical annotations identifying the inputs, outputs, and key operations required for each step, to guide development of the corresponding R scripts.
3. **Inventory documentation** – a reference record of the methodological approach taken in the national GHG Inventory.

Each source category is organised by its category code. Technical annotations appear as callout boxes beneath each step.

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1 Energy

1.1 Annex 2.1 – CO2 Emissions from Fossil Fuel Combustion

Overview. CO2 emissions from fossil fuel combustion are estimated using a bottom-up, sector-based approach aligned with the IPCC 2006 Guidelines Tier 2 method. The methodology begins with raw fuel consumption data from the Energy Information Administration (EIA), applies a series of adjustments to arrive at energy consumed for combustion purposes, and then converts adjusted consumption to CO2 emissions using fuel-specific carbon content coefficients. The primary inputs are EIA fuel consumption statistics; the primary output is CO2 emissions in MMT CO2 Eq. by fuel type and end-use sector.

1.1.1 Step 1 – Determine total fuel consumption by fuel type and sector

Fuel consumption data are obtained from the EIA in physical units and converted to energy equivalents (TBtu, using higher heating values). Data are aggregated by end-use sector (residential, commercial, industrial, transportation, electric power, and U.S. Territories) and by primary and secondary fuel type. Electricity use is allocated to sectors based on EIA retail sales to ultimate customers. For years 2010 onward, electricity consumption by electric vehicles is estimated separately and reallocated from residential and commercial sectors to transportation.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	EIA Monthly Energy Review; EIA fuel consumption surveys by sector and fuel type; EIA electricity retail sales data; Browning (2018) EV electricity methodology
Outputs	Table of unadjusted fuel consumption (TBtu) by fuel type x sector x year
Key operations	Unit conversion (physical -> TBtu); aggregation by sector and fuel type; EV electricity reallocation from residential/commercial to transportation
R notes	`dplyr::summarise()` for aggregation; conversion factors stored as named constants or lookup table; EV reallocation applied conditionally for years >= 2010
Corresponding script	`R/01_fuel_consumption_raw.R`

1.1.2 Step 2 – Subtract uses accounted for in the IPPU chapter

Portions of seven fuel types (coking coal, distillate fuel, industrial other coal, petroleum coke, natural gas, residual fuel oil, and other oil >401 degreesF) are reallocated to the Industrial Processes and Product Use (IPPU) chapter because they are consumed as raw materials rather than as combustion fuels. These amounts are removed from Energy chapter totals to avoid double-counting with IPPU. The adjustment quantities are derived from activity data used in the relevant IPPU calculations (e.g., iron and steel production, aluminum production, ammonia production, carbon black, silicon carbide, titanium dioxide, ferroalloys).

[CODING] Coding annotation – Step 2

Item	Detail
Inputs	IPPU activity data for iron & steel, aluminum, ammonia, carbon black, SiC, TiO2, ferroalloys; EIA reported coking coal consumption
Outputs	Adjustment table (TBtu) by fuel type to be subtracted from industrial sector totals
Key operations	Join IPPU-derived fuel quantities to energy consumption table; subtract adjustment amounts by fuel type
R notes	Left join on fuel type; negative values possible where IPPU consumption exceeds EIA-reported totals – handle with `pmax()` or explicit flag
Corresponding script	`R/02_ippu_adjustments.R`

1.1.3 Step 3 – Adjust for conversion of fossil fuels and exports

Three adjustments are applied. First, ethanol blended into motor gasoline is removed (biogenic carbon tracked in LULUCF, not Energy). Second, biodiesel and renewable diesel blended into distillate fuel oil are removed for the same reason (from 2009 onward). Third, industrial other coal used to produce synthetic natural gas at the Dakota Gasification Plant is adjusted: the energy content associated with CO2 exported to Canada by pipeline is subtracted from industrial other coal consumption, as that CO2 is not emitted to the U.S. atmosphere.

[CODING] Coding annotation – Step 3

Item	Detail
Inputs	EIA ethanol blending volumes; EIA biodiesel/renewable diesel volumes (2009+); Dakota Gasification Plant CO2 export volumes
Outputs	Further-adjusted consumption table with biofuel fractions and exported CO2 energy removed
Key operations	Conditional subtraction by year and fuel type; motor gasoline and distillate fuel oil adjusted separately

R notes	Year-conditional logic (`if_else()` or `case_when()`) for biodiesel adjustment start year; Dakota adjustment is a single small annual subtraction from industrial other coal
Corresponding script	`R/03_biofuel_and_conversion_adjustments.R`

1.1.4 Step 4 – Adjust sectoral allocation of distillate fuel oil and motor gasoline

EPA conducts an independent bottom-up analysis of transportation fuel consumption using Federal Highway Administration (FHWA) data. Where FHWA estimates differ from EIA's sectoral allocation, the transportation sector's distillate and motor gasoline consumption is adjusted to match the FHWA-derived figure. Because EIA national totals are considered accurate, the difference is redistributed proportionally across the residential, commercial, and industrial sectors.

[CODING] Coding annotation – Step 4

Item	Detail
Inputs	FHWA Highway Statistics (annual); EIA sectoral distillate and motor gasoline consumption
Outputs	Revised distillate and motor gasoline consumption by sector, with transportation adjusted to FHWA estimate
Key operations	Calculate difference between FHWA and EIA transportation figures; redistribute difference proportionally across residential, commercial, and industrial sectors
R notes	Proportional redistribution: non-transport sector shares calculated from unadjusted values, then applied to the adjustment delta
Corresponding script	`R/04_fhwa_sectoral_adjustment.R`

1.1.5 Step 5 – Subtract consumption for non-energy use

Fossil fuels consumed for non-energy purposes (e.g., asphalt, lubricants, plastics feedstocks) are subtracted from total fuel consumption. These fuels sequester varying fractions of their carbon content in products rather than releasing it to the atmosphere during combustion. Their emissions are estimated separately in the Carbon Emitted from Non-Energy Uses of Fossil Fuels section.

[CODING] Coding annotation – Step 5

Item	Detail
Inputs	EIA non-energy use estimates by fuel type and sector (Table A-18)
Outputs	Consumption table with non-energy fuel quantities removed
Key operations	Subtract non-energy use volumes by fuel type from total consumption
R notes	Non-energy use table joined by fuel type and year; subtraction produces the combustion-only consumption figures
Corresponding script	`R/05_non_energy_use_adjustment.R`

1.1.6 Step 6 – Subtract consumption of international bunker fuels

Emissions from international transport (aviation jet fuel, marine residual and distillate fuel oil used in international shipping) are excluded from national totals per IPCC and UNFCCC guidelines. The estimated bunker fuel volumes are subtracted from EIA transportation sector consumption figures. Bunker fuel emissions are reported separately.

[CODING] Coding annotation – Step 6

Item	Detail
Inputs	International bunker fuel estimates by fuel type (Table A-19); see also Annex 3.3 for aviation methodology
Outputs	Final adjusted consumption table (TBtu), equivalent to Tables A-5 through A-17 in the Annex
Key operations	Subtract bunker volumes from transportation sector jet fuel, residual fuel, and distillate fuel entries
R notes	Straightforward subtraction; bunker fuel table joined by fuel type and year
Corresponding script	`R/06_bunker_fuel_adjustment.R`

1.1.7 Step 7 – Determine the carbon content of all fuels

Adjusted energy consumption (TBtu) for each fuel type is multiplied by a fuel-specific carbon content coefficient (MMT C/QBtu) to yield the carbon content of combusted fuels. Coefficients are sourced from EIA and EPA and largely reflect U.S.-specific fuel qualities rather than IPCC defaults. For several fuels (motor gasoline, distillate, HGL, natural gas, crude oil, motor gasoline blending components), coefficients vary annually to reflect changes in fuel composition. For geothermal electricity generation, carbon content is estimated from net generation by geotype (flash steam, dry steam, binary) multiplied by geotype-specific coefficients.

[CODING] Coding annotation – Step 7

Item	Detail
Inputs	Adjusted fuel consumption by fuel type and sector (Steps 1-6 output); carbon content coefficient table (Table A-20), annually variable for select fuels
Outputs	Carbon content (MMT C) by fuel type and sector
Key operations	Join adjusted consumption to carbon content coefficients by fuel type and year; multiply consumption x coefficient; geothermal handled separately by geotype
R notes	Annual variability in coefficients means the join must be on both `fuel_type` and `year`; fixed coefficients can be stored as a lookup; variable ones require a time-series table
Corresponding script	`R/07_carbon_content.R`

1.1.8 Step 8 – Estimate CO2 emissions

Carbon content values are converted to CO2 emissions by applying a 100 percent oxidation fraction (per IPCC 2006 guidance). Emissions are expressed in MMT CO2 Eq. and summarised by major fuel type and consuming sector. To report emissions by final end-use sector, electric power sector emissions are distributed to residential, commercial, industrial, and transportation sectors in proportion to each sector's share of aggregate electricity consumption. Transportation emissions are further disaggregated by vehicle type and mode using fuel consumption data by vehicle type (see Annex 3.1).

[CODING] Coding annotation – Step 8

Item	Detail
Inputs	Carbon content by fuel type and sector (Step 7); electricity consumption by end-use sector (Table A-22); transportation fuel consumption by vehicle type (Annex 3.1)
Outputs	CO2 emissions (MMT CO2 Eq.) by fuel type x sector; also by final end-use sector with electric power distributed; transportation sub-allocated by vehicle/mode
Key operations	Multiply C content x (44/12) to convert C to CO2; allocate electric power emissions to end-use sectors by electricity share; join transport mode allocation factors
R notes	C-to-CO2 conversion factor is 44.01/12.011 ~ 3.664; electric power allocation is a weighted redistribution; final output table structure should mirror Annex Tables A-5 through A-17
Corresponding script	`R/08_co2_emissions.R`

[NOTE] Summary and open questions

This is a Tier 2 bottom-up methodology using EIA aggregate fuel consumption as its foundation. Key assumptions include: 100% oxidation of carbon during combustion; HHV (not LHV) energy basis throughout; use of No. 2 distillate as representative of all distillate grades; and U.S.-specific carbon content coefficients that vary annually for several fuel types.

- **Open question:** EV electricity reallocation methodology (Browning 2018) – confirm the current implementation aligns with the most recent technical memo.
- **Open question:** IPPU adjustment quantities (Step 2) need to be sourced consistently from the same IPPU module outputs to ensure cross-chapter consistency.
- **Open question:** Confirm treatment of geothermal in Step 7 – binary/flash hybrids may require special handling.

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2 Industrial Processes and Product Use (IPPU)

2.1 4.1 Cement Production

Overview. CO2 emissions from cement production arise from the calcination of calcium carbonate (limestone) during clinker production. The methodology applies a Tier 2 approach: national clinker production data are multiplied by a fixed emission factor (0.510 metric tons CO2 per metric ton clinker), with a 2% correction factor applied to account for cement kiln dust (CKD) not returned to the kiln. Data source transitions from USGS (1990-2013) to GHGRP (2014-2022).

2.1.1 Step 1 – Obtain clinker production data

National clinker production (metric tons) is compiled. For 1990-2013, data come from USGS mineral commodity surveys. From 2014 onward, GHGRP Subpart H facility-level reports are aggregated to a national total. An overlap comparison is performed to check consistency between the two sources at the transition year.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	USGS clinker production data (1990-2013); EPA GHGRP Subpart H (2014-2022)
Outputs	Annual clinker production time series (metric tons), 1990-2022
Key operations	Aggregate GHGRP facility-level data to national; splice USGS and GHGRP series at 2013/2014; overlap check
R notes	GHGRP data likely imported from CSV/API; `dplyr::summarise()` for national rollup; splice with `bind_rows()` after filtering by year range
Corresponding script	`R/cement/01_clinker_production.R`

2.1.2 Step 2 – Apply clinker emission factor and CKD correction

Clinker production is multiplied by the IPCC Tier 2 emission factor of 0.510 metric tons CO2 per metric ton clinker. A CKD correction factor of 1.02 is then applied to account for CO2 emissions from calcinated CKD that is not returned to the kiln.

Emissions = clinker production x 0.510 x 1.02

[CODING] Coding annotation – Step 2

Item	Detail
Inputs	Clinker production time series (Step 1); fixed emission factor = 0.510 t CO2/t clinker; CKD correction factor = 1.02
Outputs	Annual CO2 emissions from cement production (MMT CO2 Eq.)
Key operations	Multiply production x EF x CKD correction; unit conversion to MMT
R notes	Both factors are constants – store in a `params` list or header section for transparency; one-liner mutate
Corresponding script	`R/cement/02_emissions_calculation.R`

[NOTE] Summary and open questions

This is a straightforward Tier 2 calculation with two fixed parameters. The main sources of uncertainty are the assumed 65% lime fraction in clinker and the default CKD correction factor.

- **Open question:** Confirm that the GHGRP Subpart H data used does not include fuel combustion CO2 (only process CO2); some facilities report combined totals via CEMS.

2.2 4.2 Lime Production

Overview. CO2 emissions from lime production result from the calcination of calcium and magnesium carbonates. The Tier 2 methodology multiplies high-calcium and dolomitic lime production separately by their respective stoichiometric emission factors, derived from the CaO and MgO content of lime (assumed 95% for both types).

2.2.1 Step 1 – Obtain lime production data by type

Annual production of high-calcium lime and dolomitic lime (metric tons) is obtained from USGS mineral commodity data for 1990-2021, and from GHGRP Subpart S for more recent years where applicable.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	USGS lime production by type; GHGRP Subpart S facility data (where used)
Outputs	Annual production of high-calcium and dolomitic lime (metric tons) by year
Key operations	Separate production into high-calcium vs. dolomitic categories; aggregate to national
R notes	Two-row output per year (one per lime type); or wide format with two columns
Corresponding script	`R/lime/01_lime_production.R`

2.2.2 Step 2 – Apply type-specific emission factors

Each lime type is multiplied by its stoichiometric emission factor:

- High-calcium lime: EF = 0.95 x (44.01/56.08) = 0.7454 metric tons CO2/metric ton lime
- Dolomitic lime: EF = 0.95 x [(44.01/56.08) + (44.01/84.31)] x 0.5 ~ 0.8364 metric tons CO2/metric ton lime

Total emissions are the sum across lime types.

[CODING] Coding annotation – Step 2

Item	Detail
Inputs	Lime production by type (Step 1); type-specific emission factors (derived from stoichiometry)
Outputs	CO2 emissions (MMT CO2 Eq.) by lime type and total
Key operations	Multiply production x EF by lime type; sum to total
R notes	Emission factors are constants derived from molecular weights – document derivation clearly; `group_by(lime_type)` then `summarise()`
Corresponding script	`R/lime/02_emissions_calculation.R`

[NOTE] Summary and open questions

- **Open question:** Confirm current data source for lime production – USGS coverage vs. GHGRP Subpart S and transition year if applicable.

2.3 4.3 Glass Production

Overview. CO2 emissions from glass production arise from the calcination of carbonate inputs (limestone, dolomite, soda ash, and other carbonates). A Tier 3 approach is used: quantities of each carbonate input are multiplied by carbonate-specific emission factors and mass fractions. For 2010-2022, GHGRP data provide carbonate input quantities; USGS data on soda ash supplement this. Pre-2010 data use a different source blend.

2.3.1 Step 1 – Obtain carbonate input quantities by type

For 2010-2022: quantities of limestone, dolomite, soda ash, barium carbonate, potassium carbonate, lithium carbonate, and strontium carbonate used in glass production are obtained from GHGRP and USGS. For earlier years, alternative data sources are used.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	GHGRP facility-level carbonate inputs (2010-2022); USGS soda ash consumption for glass; pre-2010 sources
Outputs	Annual carbonate input quantities (metric tons) by carbonate type
Key operations	Aggregate GHGRP facility data by carbonate type; supplement with USGS soda ash; handle pre-2010 differently
R notes	Year-conditional data sourcing – consider using a `targets` pipeline with separate data ingestion targets per source/period
Corresponding script	`R/glass/01_carbonate_inputs.R`

2.3.2 Step 2 – Apply carbonate-specific emission factors

Each carbonate is multiplied by its stoichiometric CO2 emission factor (metric tons CO2 per metric ton carbonate) and its average mineral mass fraction. Total emissions are summed across all carbonate types.

Emissions = Sum (carbonate quantity x emission factor x mass fraction)

[CODING] Coding annotation – Step 2

Item	Detail
Inputs	Carbonate quantities by type (Step 1); IPCC-derived emission factors and mass fractions by carbonate type
Outputs	CO2 emissions (MMT CO2 Eq.) from glass production
Key operations	Join carbonate quantities to EF lookup; multiply and sum
R notes	EF lookup table keyed on carbonate type; `left_join()` then `mutate()` and `summarise()`
Corresponding script	`R/glass/02_emissions_calculation.R`

[NOTE] Summary and open questions

- **Open question:** Clarify which specific carbonate types are covered by GHGRP Subpart H (glass) versus those requiring USGS supplementation. Confirm whether "other carbonates" in GHGRP align with the Inventory categorisation.

2.4 4.4 Other Process Uses of Carbonates

Overview. This category covers CO2 emissions from limestone and dolomite consumed in non-glass, non-cement applications – flux stone, flue gas desulfurization (FGD), chemical stone, mine dusting, acid water treatment, and acid neutralization. It also covers soda ash consumption not associated with glass manufacturing. A Tier 2 approach is used for limestone and dolomite; a separate approach is used for soda ash.

2.4.1 Step 1 – Obtain limestone and dolomite consumption by end use

Annual consumption quantities (metric tons) of limestone and dolomite for each applicable end use are obtained from USGS.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	USGS mineral commodity data: limestone and dolomite consumption by end use
Outputs	Annual consumption by carbonate type and end use
Key operations	Filter to applicable end uses; aggregate if needed
R notes	Straightforward read and filter
Corresponding script	`R/carbonates/01_consumption_data.R`

2.4.2 Step 2 – Apply emission factors for limestone and dolomite

Fixed stoichiometric emission factors are applied:

- Limestone: 0.43971 metric tons CO2 per metric ton carbonate
- Dolomite: 0.47732 metric tons CO2 per metric ton carbonate

[CODING] Coding annotation – Step 2

Item	Detail
Inputs	Consumption by type (Step 1); fixed emission factors
Outputs	CO2 emissions from limestone and dolomite consumption
Key operations	Multiply consumption x EF by carbonate type; sum
R notes	Constants stored in params; one mutate + summarise
Corresponding script	`R/carbonates/02_limestone_dolomite_emissions.R`

2.4.3 Step 3 – Estimate emissions from soda ash consumption

Soda ash consumption for non-glass uses is obtained from USGS. A soda ash-specific emission factor is applied, derived from the stoichiometry of Na2CO3 decomposition (consistent with IPCC 2006 Section 2.4.2.1).

[CODING] Coding annotation – Step 3

Item	Detail
Inputs	USGS soda ash consumption (non-glass); stoichiometric emission factor for Na2CO3
Outputs	CO2 emissions from soda ash consumption
Key operations	Multiply consumption x EF
R notes	Can be combined with limestone/dolomite output in a single emissions table
Corresponding script	`R/carbonates/03_soda_ash_emissions.R`

[NOTE] Summary and open questions

- **Open question:** Confirm whether flux stone used in iron and steel production is excluded here (should be captured under 4.18) to avoid double-counting.

2.5 4.5 Ammonia Production

Overview. CO2 emissions from ammonia production arise from the use of natural gas or petroleum coke as a feedstock (hydrogen source) in steam methane reforming. For 2010-2022, a Tier 3 approach uses process CO2 emissions reported directly to GHGRP Subpart G. For 1990-2009, a Tier 2 approach uses feedstock consumption data multiplied by emission factors.

2.5.1 Step 1 – Obtain process CO2 data (2010-2022)

Facility-level CO2 emissions from ammonia production are obtained from GHGRP Subpart G and aggregated to a national annual total.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	GHGRP Subpart G facility reports (2010-2022)
Outputs	National CO2 emissions from ammonia production, 2010-2022
Key operations	Aggregate facility-level emissions to national total
R notes	`dplyr::summarise()` on GHGRP data filtered to Subpart G and ammonia production
Corresponding script	`R/ammonia/01_ghgrp_emissions.R`

2.5.2 Step 2 – Estimate emissions from feedstock data (1990-2009)

For years prior to GHGRP, feedstock consumption (natural gas and petroleum coke) used for ammonia production is obtained from EIA and other sources. Emissions are estimated by multiplying feedstock quantities by their carbon content coefficients (consistent with Tier 2 mass balance).

[CODING] Coding annotation – Step 2

Item	Detail
Inputs	EIA natural gas and petroleum coke feedstock data for ammonia plants (1990-2009); carbon content coefficients from Annex 2.2
Outputs	CO2 emissions from ammonia production, 1990-2009
Key operations	Feedstock consumption x carbon content coefficient; convert to CO2
R notes	Share carbon content coefficients with the Energy chapter module for consistency
Corresponding script	`R/ammonia/02_pre2010_emissions.R`

2.5.3 Step 3 – Splice time series

The pre-2010 (Tier 2) and post-2010 (Tier 3/GHGRP) estimates are combined into a single consistent time series.

[CODING] Coding annotation – Step 3

Item	Detail
Inputs	Outputs from Steps 1 and 2
Outputs	Complete 1990-2022 CO2 emissions time series for ammonia production
Key operations	`bind_rows()` with year filter to avoid overlap; overlap-year comparison for QC
Corresponding script	`R/ammonia/03_splice_time_series.R`

[NOTE] Summary and open questions

- **Open question:** Feedstock CO2 from ammonia production is excluded from the Energy chapter (Step 2 of Annex 2.1). Confirm the exact adjustment amounts reconcile with the IPPU estimate for consistency.

2.6 4.6 Urea Consumption for Non-Agricultural Purposes

Overview. When urea is applied or used industrially (non-fertiliser), the CO2 incorporated during its synthesis is released. Emissions are estimated by multiplying non-agricultural urea consumption by a stoichiometric factor (44/60), representing the mass of CO2 relative to urea (CH4N2O, MW = 60; CO2, MW = 44).

2.6.1 Step 1 – Obtain non-agricultural urea consumption

Annual urea consumption for non-agricultural purposes (metric tons) is obtained from industry sources and USGS.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	Non-agricultural urea consumption data (USGS or industry)

Outputs	Annual non-agricultural urea consumption (metric tons)
Corresponding script	`R/urea/01_consumption.R`

2.6.2 Step 2 – Apply stoichiometric emission factor

Emissions = urea consumption x (44/60)

[CODING] Coding annotation – Step 2

Item	Detail
Inputs	Non-agricultural urea consumption (Step 1)
Outputs	CO2 emissions (MMT CO2 Eq.) from urea use
Key operations	Multiply consumption x 44/60; unit conversion to MMT
R notes	Single constant factor; straightforward mutate
Corresponding script	`R/urea/02_emissions_calculation.R`

2.7 4.7 Nitric Acid Production

Overview. N2O is emitted as a byproduct during the catalytic oxidation of ammonia in nitric acid production. For 2010-2022, a Tier 3 approach uses N2O emissions and production data reported directly to GHGRP. For 1990-2009, a Tier 2 approach uses U.S. Census Bureau production data multiplied by IPCC default or abatement-adjusted emission factors.

2.7.1 Step 1 – Obtain N2O emissions and production data (2010-2022)

Facility-level N2O emissions and nitric acid production volumes are obtained from GHGRP and aggregated nationally. All U.S. facilities producing weak nitric acid (30-70%) are required to report.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	GHGRP Subpart V facility-level N2O emissions and nitric acid production (2010-2022)
Outputs	National N2O emissions and production, 2010-2022
Key operations	Aggregate facility reports to national total
Corresponding script	`R/nitric_acid/01_ghgrp_data.R`

2.7.2 Step 2 – Estimate emissions from production data (1990-2009)

Nitric acid production from U.S. Census Bureau data is multiplied by Tier 2 emission factors, accounting for installed abatement technology where applicable.

[CODING] Coding annotation – Step 2

Item	Detail
Inputs	U.S. Census Bureau nitric acid production (1990-2009); IPCC default EFs; abatement technology assumptions
Outputs	N2O emissions, 1990-2009
Key operations	Production x EF; adjust for abatement where applicable
Corresponding script	`R/nitric_acid/02_pre2010_emissions.R`

2.7.3 Step 3 – Splice and express as CO2 equivalents

The two time series are combined and N2O emissions are converted to CO2 equivalents using the applicable GWP.

[CODING] Coding annotation – Step 3

Item	Detail
Key operations	`bind_rows()`; multiply N2O (metric tons) x GWP to get MMT CO2 Eq.

R notes	GWP value should be stored as a named constant consistent with the GWP set used across all Inventory categories
Corresponding script	`R/nitric_acid/03_splice_and_gwp.R`

2.8 4.8 Adipic Acid Production

Overview. N2O is produced as a byproduct of the catalytic oxidation of cyclohexane or cyclohexanone/cyclohexanol during adipic acid production. The Tier 2 methodology uses production data with IPCC default emission factors, adjusted for abatement where applicable. GHGRP data are used where available.

2.8.1 Step 1 – Obtain adipic acid production data

Annual adipic acid production (metric tons) is obtained from U.S. Census Bureau and/or GHGRP data.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	U.S. Census Bureau or GHGRP production data
Outputs	Annual adipic acid production (metric tons)
Corresponding script	`R/adipic_acid/01_production.R`

2.8.2 Step 2 – Apply emission factor with abatement adjustment

Unadjusted N2O emissions are estimated using a default emission factor. Where abatement systems (catalytic decomposition or thermal oxidation) are installed, an abatement-adjusted effective emission factor is applied.

Emissions = production x EF x (1 - abatement efficiency x abatement fraction)

[CODING] Coding annotation – Step 2

Item	Detail
Inputs	Production (Step 1); IPCC default EF; plant-level abatement data
Outputs	N2O emissions (MMT CO2 Eq.) from adipic acid production
Key operations	EF x production; abatement adjustment where applicable; GWP conversion
Corresponding script	`R/adipic_acid/02_emissions_calculation.R`

2.9 4.9 Caprolactam, Glyoxal and Glyoxylic Acid Production

Overview. N2O is generated during the high-temperature catalytic oxidation of ammonia in caprolactam production. Emissions are calculated using the IPCC Tier 1 method: caprolactam production is multiplied by a default emission factor. Glyoxal and glyoxylic acid are minor contributors also covered under this section.

2.9.1 Step 1 – Obtain production data

Annual caprolactam, glyoxal, and glyoxylic acid production (metric tons) from U.S. Census Bureau.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	U.S. Census Bureau production data
Outputs	Annual production by chemical
Corresponding script	`R/caprolactam/01_production.R`

2.9.2 Step 2 – Apply Tier 1 emission factors

N2O emissions = production x default EF (IPCC 2006 Equation 3.9). Convert to CO2 equivalents using GWP.

[CODING] Coding annotation – Step 2

Item	Detail
Inputs	Production by chemical; IPCC Tier 1 EFs
Outputs	N2O emissions (MMT CO2 Eq.)
Key operations	Production x EF x GWP; sum across chemicals
Corresponding script	`R/caprolactam/02_emissions_calculation.R`

2.10 4.10 Carbide Production and Consumption

Overview. CO2 and CH4 are emitted during the production of silicon carbide (SiC). CO2 is also emitted from consumption of SiC as a reducing agent. Production emissions use IPCC Tier 1 default factors. Consumption emissions use a country-specific stoichiometric approach (no IPCC guidance exists for SiC consumption).

2.10.1 Step 1 – Apply Tier 1 emission factors to SiC production

Annual SiC production (metric tons) x IPCC default EFs:

- CO2: 2.62 metric tons CO2 per metric ton SiC
- CH4: IPCC default CH4 factor

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	U.S. Census Bureau SiC production; IPCC default EFs
Outputs	CO2 and CH4 emissions from SiC production
Corresponding script	`R/carbide/01_production_emissions.R`

2.10.2 Step 2 – Estimate CO2 from SiC consumption

SiC consumption data (metric tons) are multiplied by a stoichiometric emission factor derived from the chemistry of SiC use as a reducing agent.

[CODING] Coding annotation – Step 2

Item	Detail
Inputs	SiC consumption data; country-specific stoichiometric EF
Outputs	CO_2 emissions from SiC consumption
Corresponding script	`R/carbide/02_consumption_emissions.R`

2.11 4.11 Titanium Dioxide Production

Overview. CO_2 is emitted from the chloride process of TiO_2 production, where petroleum coke is used as a reducing agent. Emissions are calculated using IPCC Tier 1 methodology: annual TiO_2 production multiplied by a chloride-process-specific emission factor. The petroleum coke portion of these emissions is adjusted out of the Energy chapter to avoid double-counting.

2.11.1 Step 1 – Obtain TiO2 production data and apply emission factor

Annual TiO_2 production (metric tons) from USGS. Multiply by IPCC chloride process emission factor.

Emissions = TiO_2 production x EF (chloride process)

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	USGS TiO_2 production; IPCC Tier 1 EF for chloride process
Outputs	CO_2 emissions (MMT CO_2 Eq.) from TiO_2 production
Key operations	Single multiply; unit conversion
R notes	Confirm petroleum coke adjustment to Energy chapter is handled in Energy module (Step 2 of Annex 2.1)
Corresponding script	`R/tio2/01_emissions_calculation.R`

2.12 4.12 Soda Ash Production

Overview. CO_2 is emitted during the calcination of trona ore to produce soda ash (Solvay-equivalent natural process). The IPCC Tier 1 default emission factor of 0.0974 metric tons CO_2 per metric ton of trona ore is applied to annual trona mining/consumption data.

2.12.1 Step 1 – Obtain trona ore consumption and apply emission factor

Annual trona ore consumed for soda ash production (metric tons) from USGS. Multiply by IPCC Tier 1 EF.

Emissions = trona consumption x 0.0974

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	USGS trona ore production/consumption data; IPCC Tier 1 EF = 0.0974 t CO_2/t trona
Outputs	CO_2 emissions (MMT CO_2 Eq.) from soda ash production
Key operations	Single multiply; unit conversion
Corresponding script	`R/soda_ash/01_emissions_calculation.R`

2.13 4.13 Petrochemical Production

Overview. CO_2 and CH_4 are emitted during the production of carbon black, ethylene, ethylene dichloride, ethylene oxide, methanol, and acrylonitrile. A Tier 2 total feedstock carbon mass balance approach is used for CO_2 from most chemicals (carbon black, ethylene, ethylene dichloride, ethylene oxide, methanol); GHGRP data are used where available. Acrylonitrile uses IPCC Tier 1 defaults. CH_4 emissions are estimated separately using IPCC Tier 1 methods.

2.13.1 Step 1 – Obtain production data by chemical

Annual production (metric tons) for each covered petrochemical from U.S. Census Bureau and GHGRP.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	U.S. Census Bureau chemical production data; GHGRP facility-level data where applicable
Outputs	Annual production by chemical (metric tons)
Corresponding script	`R/petrochemicals/01_production.R`

2.13.2 Step 2 – Apply chemical-specific emission factors for CO2

Multiply each chemical's production by its specific CO2 emission factor:

- Carbon black: 2.59 t CO2/t produced
- Ethylene: 0.79 t CO2/t produced
- Ethylene dichloride: 0.040 t CO2/t produced
- Ethylene oxide: 0.46 t CO2/t produced
- Acrylonitrile: 1.00 t CO2/t produced (Tier 1 default)

[CODING] Coding annotation – Step 2

Item	Detail
Inputs	Production by chemical (Step 1); chemical-specific EF lookup table
Outputs	CO2 emissions by chemical and total
Key operations	Join production to EF table; multiply; sum
R notes	EF lookup keyed on chemical name; `left_join()` then `mutate()`
Corresponding script	`R/petrochemicals/02_co2_emissions.R`

2.13.3 Step 3 – Apply Tier 1 factors for CH4

CH4 is estimated using IPCC Tier 1 default factors for acrylonitrile (0.18 kg CH4/t produced) and other applicable chemicals.

[CODING] Coding annotation – Step 3

Item	Detail
Inputs	Production data; IPCC Tier 1 CH4 EFs
Outputs	CH4 emissions (kt); CO2 Eq. after GWP conversion
Corresponding script	`R/petrochemicals/03_ch4_emissions.R`

2.14 4.14 HCFC-22 Production

Overview. HFC-23 (a potent GHG) is generated as a byproduct during the synthesis of HCFC-22. Tier 3 methods (facility-specific data) are used for five of the eight plants that have operated since 1990. GHGRP Subpart O provides data for 2010-2022; voluntary industry reporting covers 1990-2009.

2.14.1 Step 1 – Obtain facility-level HFC-23 emissions data

GHGRP Subpart O (2010-2022) and voluntarily reported data from industry (1990-2009) are compiled and aggregated to a national total.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	GHGRP Subpart O; voluntary industry reports (1990-2009)
Outputs	National HFC-23 emissions (metric tons) from HCFC-22 production, 1990-2022
Key operations	Aggregate facility data; splice voluntary and GHGRP series
Corresponding script	`R/hcfc22/01_hfc23_emissions.R`

2.14.2 Step 2 – Convert to CO2 equivalents

HFC-23 emissions (metric tons) x GWP -> MMT CO2 Eq.

[CODING] Coding annotation – Step 2

Item	Detail
Key operations	Multiply by GWP for HFC-23; unit conversion to MMT
R notes	GWP stored as named constant; confirm correct AR GWP set is used
Corresponding script	`R/hcfc22/02_gwp_conversion.R`

2.15 4.15 Production of Fluorochemicals Other Than HCFC-22

Overview. This category covers fluorinated GHG emissions from production of HFCs, PFCs, SF6, NF3, and other fluorinated compounds at U.S. facilities. The Tier 3 method uses facility-specific process vent and fugitive emission data from GHGRP Subpart L (2011-2022). For 1990-2010, a combination of Tier 1/2 methods and voluntary reporting is used.

2.15.1 Step 1 – Obtain facility-level fluorinated GHG emissions

GHGRP Subpart L data (2011-2022) are aggregated by gas and compound. For prior years, Tier 1 default factors or voluntary data are used.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	GHGRP Subpart L; voluntary/Tier 1 estimates for pre-2011
Outputs	Fluorinated GHG emissions by compound (metric tons)
Corresponding script	`R/fluorochemicals/01_emissions.R`

2.15.2 Step 2 – Convert to CO2 equivalents

Each fluorinated gas x compound-specific GWP -> MMT CO2 Eq. Sum across compounds.

[CODING] Coding annotation – Step 2

Item	Detail
Inputs	Emissions by compound; GWP lookup table
Outputs	Total fluorinated GHG emissions (MMT CO2 Eq.)
R notes	GWP table keyed on compound name; join and multiply
Corresponding script	`R/fluorochemicals/02_gwp_conversion.R`

2.16 4.16 Carbon Dioxide Consumption

Overview. CO2 extracted from natural reservoirs or captured from industrial processes and used for non-EOR commercial applications (e.g., food and beverage, chemical manufacturing) is eventually released to the atmosphere. Emissions are estimated using GHGRP Subpart PP facility reports (2010-2022); a country-specific Tier 3 approach. Pre-2010 estimates use alternative data sources.

2.16.1 Step 1 – Obtain CO2 extraction and transfer data

GHGRP Subpart PP data on CO2 extracted and transferred to non-EOR end uses are aggregated nationally.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	GHGRP Subpart PP facility reports (2010-2022)
Outputs	National CO2 consumed for non-EOR purposes (metric tons)
Corresponding script	`R/co2_consumption/01_ghgrp_data.R`

2.16.2 Step 2 – Assume full atmospheric release

CO2 consumed for non-EOR purposes is assumed to be fully and immediately released to the atmosphere.

Emissions = CO2 consumed (metric tons), converted to MMT CO2 Eq.

[CODING] Coding annotation – Step 2

Item	Detail
Key operations	Unit conversion only (metric tons -> MMT); no EF needed
Corresponding script	`R/co2_consumption/02_emissions.R`

2.17 4.17 Phosphoric Acid Production

Overview. CO2 is released when inorganic carbon (calcium carbonate) in phosphate rock reacts during wet-process phosphoric acid production. A country-specific methodology (consistent with Tier 1) multiplies the inorganic carbon content of phosphate rock by the quantity consumed, accounting for domestic production and net imports.

2.17.1 Step 1 – Obtain phosphate rock consumption data

Annual phosphate rock used for phosphoric acid production (metric tons) from USGS, accounting for domestic production and net imports.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	USGS phosphate rock data; import/export data for net consumption
Outputs	Annual phosphate rock consumption (metric tons)
Corresponding script	`R/phosphoric_acid/01_phosphate_rock.R`

2.17.2 Step 2 – Apply inorganic carbon content and calculate emissions

Average inorganic carbon content (as CO2 fraction) of U.S. phosphate rock is multiplied by the quantity consumed.

Emissions = rock consumption x average CaCO3 fraction x EF (CO2/CaCO3)

[CODING] Coding annotation – Step 2

Item	Detail
Inputs	Phosphate rock consumption (Step 1); average $CaCO_3$ content of U.S. phosphate rock; stoichiometric EF
Outputs	CO_2 emissions (MMT CO_2 Eq.)
Corresponding script	`R/phosphoric_acid/02_emissions_calculation.R`

2.18 4.18 Iron and Steel Production

Overview. CO_2 is emitted from iron and steel production through: (1) metallurgical coke production, (2) pig iron production in blast furnaces, (3) basic oxygen furnace (BOF) steel production, (4) electric arc furnace (EAF) steel production, and (5) sinter and pellet production. All processes use a country-specific Tier 2 carbon mass balance approach. Carbonaceous inputs (coke, coal, limestone, electrodes, etc.) are tracked; non-CO_2 carbon outputs (pig iron, steel, CO in off-gases) are subtracted.

2.18.1 Step 1 – Compile carbonaceous input and output data

Annual quantities of all carbon-containing inputs and outputs for each process stage (coke ovens, blast furnaces, BOF, EAF) from USGS, AISI, and GHGRP.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	USGS iron and steel mineral commodity data; AISI annual statistical reports; GHGRP Subpart Q
Outputs	Mass-balance tables of carbon inputs and outputs by process stage and year
Key operations	Compile inputs and outputs by process; calculate net carbon released
R notes	Complex multi-process mass balance – consider one R script per process (coke, blast furnace, BOF, EAF), with a consolidation script
Corresponding script	`R/iron_steel/01_mass_balance_data.R`

2.18.2 Step 2 – Calculate CO2 from each process via mass balance

For each process: CO2 = (total carbon in inputs - carbon retained in products/by-products) x (44/12). Processes where Tier 1 defaults are used (sinter, pellets, DRI) apply IPCC default factors to production volumes instead.

[CODING] Coding annotation – Step 2

Item	Detail
Inputs	Mass balance tables (Step 1); Tier 1 EFs for sinter/pellet/DRI production
Outputs	CO2 emissions by process and total (MMT CO2 Eq.)
Key operations	Net carbon x (44/12) per process; Tier 1 production x EF for remaining processes; sum
Corresponding script	`R/iron_steel/02_emissions_calculation.R`

[NOTE] Summary and open questions

- **Open question:** Significant 2020-2022 activity data were not available; these years use 2019 values adjusted with GHGRP emissions data. Flag for update in future iterations.
- **Open question:** Coking coal and other fuels used in iron and steel production are adjusted out of the Energy chapter (Step 2, Annex 2.1). Confirm reconciliation between the Energy chapter adjustment and the IPPU mass balance quantities.

2.19 4.19 Ferroalloy Production

Overview. CO2 and CH4 are emitted during the reduction of metal oxides in electric arc furnaces using carbonaceous reducing agents (coke, coal). The Tier 2 methodology applies a carbon mass balance: carbon inputs to each ferroalloy process minus carbon retained in the product. IPCC Tier 2 methods are used.

2.19.1 Step 1 – Obtain ferroalloy production data and carbon inputs

Annual production by ferroalloy type and associated carbonaceous input quantities from USGS.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	USGS ferroalloy production and carbon input data
Outputs	Production and carbon input by ferroalloy type and year
Corresponding script	`R/ferroalloys/01_production_data.R`

2.19.2 Step 2 – Apply carbon mass balance

CO2 = (carbon in - carbon retained in alloy) x (44/12). CH4 estimated separately via IPCC default factor.

[CODING] Coding annotation – Step 2

Item	Detail
Inputs	Carbon inputs/outputs by alloy type; IPCC CH4 default EF
Outputs	CO2 and CH4 emissions (MMT CO2 Eq.)
Corresponding script	`R/ferroalloys/02_emissions_calculation.R`

2.20 4.20 Aluminum Production

Overview. CO2 (from anode consumption) and PFCs (CF4 and C2F6, from anode effects) are emitted during primary aluminum smelting. For 2010-2022, GHGRP Subpart F provides facility-specific process emissions. For earlier years, IPCC Tier 2 methods or voluntary industry data are used. The petroleum coke consumed in anode production is excluded from the Energy chapter non-energy use calculations.

2.20.1 Step 1 – Obtain process emissions data (2010-2022)

Aggregate GHGRP Subpart F facility reports: CO2 from anode consumption (prebake and Soderberg cells), CO2 from anode baking, and PFC emissions from anode effects.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	GHGRP Subpart F: CF4, C2F6, and CO2 by facility
Outputs	National CO2 and PFC emissions from primary aluminum, 2010-2022
Corresponding script	`R/aluminum/01_ghgrp_emissions.R`

2.20.2 Step 2 – Estimate pre-2010 emissions

Use IPCC Tier 2 methods (aluminum production x carbon mass balance for CO2; anode effect parameters for PFCs) and voluntary industry reporting where available.

[CODING] Coding annotation – Step 2

Item	Detail
Inputs	USGS aluminum production; IPCC Tier 2 EFs; voluntary industry data
Outputs	CO2 and PFC emissions, 1990-2009
Corresponding script	`R/aluminum/02_pre2010_emissions.R`

2.20.3 Step 3 – Convert PFCs to CO2 equivalents and combine

CF4 and C2F6 emissions x respective GWPs -> MMT CO2 Eq. Add to CO2 emissions and splice time series.

[CODING] Coding annotation – Step 3

Item	Detail
Key operations	GWP conversions for CF4 and C2F6; `bind_rows()` to splice series
Corresponding script	`R/aluminum/03_gwp_and_splice.R`

2.21 4.21 Magnesium Production and Processing

Overview. SF6 (and increasingly SO2 and HFC-134a) is used as a cover gas to prevent oxidation of molten magnesium during primary production, secondary production (recycling), and die casting. SF6 consumption is assumed equivalent to emissions. Data come from EPA's SF6 Emission Reduction Partnership (1999-2010) and GHGRP Subpart T (2010-2022).

2.21.1 Step 1 – Obtain SF6 consumption/emissions data

Partnership-reported SF6 consumption (1999-2010) and GHGRP Subpart T data (2010-2022) are compiled by production/processing category. Pre-1999 years use modelled or interpolated estimates.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	EPA SF6 Partnership reports; GHGRP Subpart T
Outputs	Annual SF6 emissions (metric tons) by category
Corresponding script	`R/magnesium/01_sf6_emissions.R`

2.21.2 Step 2 – Convert to CO2 equivalents

SF6 emissions x GWP -> MMT CO2 Eq.

[CODING] Coding annotation – Step 2

Item	Detail
Key operations	Multiply by SF6 GWP; unit conversion
Corresponding script	`R/magnesium/02_gwp_conversion.R`

2.22 4.22 Lead Production

Overview. CO2 is emitted during smelting of lead using direct smelting (primary) and secondary (recycled) production routes. The IPCC Tier 1 method applies separate emission factors for each production route and sums the results.

Emissions = (direct smelting production x EF_DS) + (secondary production x EF_S)

2.22.1 Step 1 – Obtain lead production by route and apply emission factors

Annual lead production by production route (metric tons) from USGS. Apply IPCC Tier 1 EFs.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	USGS lead production by route; IPCC Tier 1 EFs for direct smelting and secondary production
Outputs	CO2 emissions (MMT CO2 Eq.) from lead production
Key operations	Two-term calculation; join production by route to EF; sum
Corresponding script	`R/lead/01_emissions_calculation.R`

2.23 4.23 Zinc Production

Overview. CO2 is emitted from zinc production using electrothermic (primary) and Waelz kiln (secondary) processes. The IPCC Tier 1 method applies a single default emission factor (weighted average across process types) to total zinc production.

Emissions = zinc production x EF_default

2.23.1 Step 1 – Obtain zinc production data and apply emission factor

Annual zinc production (metric tons) from USGS. Multiply by IPCC Tier 1 default EF.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	USGS zinc production; IPCC Tier 1 default EF
Outputs	CO_2 emissions (MMT CO_2 Eq.) from zinc production
Key operations	Single multiply; unit conversion
Corresponding script	`R/zinc/01_emissions_calculation.R`

2.24 4.24 Electronics Industry

Overview. Fluorinated GHGs (HFCs, PFCs, SF_6, NF_3, and others) are used in semiconductor, flat panel display, and photovoltaic manufacturing for plasma etching and chemical vapour deposition chamber cleaning. Emissions are estimated using a combination of GHGRP Subpart I data, EPA Partnership voluntary reports, and the PFC Emissions Vintage Model (PEVM). Methodological approach varies by sub-sector and time period.

2.24.1 Step 1 – Obtain fluorinated GHG emissions by sub-sector and period

GHGRP Subpart I data (semiconductors, flat panels, PVs) are compiled for applicable years. Partnership data and PEVM model outputs supplement or replace GHGRP data for earlier periods or non-reporting companies.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	GHGRP Subpart I; EPA PFC Partnership data; PEVM model outputs; industry activity data (manufactured layer area, capacity)
Outputs	Fluorinated GHG emissions (metric tons) by compound and sub-sector
R notes	Complex multi-source, multi-period assembly – recommend one ingestion script per data source, then a consolidation/splice script
Corresponding script	`R/electronics/01_emissions_assembly.R`

2.24.2 Step 2 – Convert to CO2 equivalents

Each fluorinated compound x compound-specific GWP -> MMT CO2 Eq. Sum across compounds and sub-sectors.

[CODING] Coding annotation – Step 2

Item	Detail
Inputs	Emissions by compound; GWP lookup table
Outputs	Total electronics industry fluorinated GHG emissions (MMT CO2 Eq.)
Corresponding script	`R/electronics/02_gwp_conversion.R`

2.25 4.25 Substitution of Ozone Depleting Substances

Overview. HFCs, PFCs, and CO2 are used as replacements for ozone depleting substances (ODS) in refrigeration, air conditioning, foam blowing, fire suppression, aerosols, and other applications. Emissions are estimated using EPA's Vintaging Model, a Tier 2 approach that tracks annual equipment "vintages," applies equipment-specific charge sizes and leak rates, and models cumulative emissions from the installed base of equipment over its lifetime.

2.25.1 Step 1 – Model equipment stock and annual emissions via Vintaging Model

The Vintaging Model estimates the quantity of new equipment placed into service each year, its ODS-substitute chemical charge, and annual leakage and end-of-life emission rates. Emissions for each equipment vintage and compound are accumulated across years.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	Vintaging Model outputs by end-use, compound, and year; market penetration data; equipment population estimates
Outputs	Annual emissions by compound and end-use (metric tons)

R notes	If the Vintaging Model output is a fixed dataset (CSV/Excel), ingest and reshape; if rerunnable, wrap in a targets pipeline step
Corresponding script	`R/ods_substitutes/01_vintaging_model_output.R`

2.25.2 Step 2 – Convert to CO2 equivalents

Each compound x GWP -> MMT CO2 Eq. Sum across end-uses and compounds.

[CODING] Coding annotation – Step 2

Item	Detail
Inputs	Emissions by compound; GWP lookup
Outputs	Total ODS substitute emissions (MMT CO2 Eq.)
Corresponding script	`R/ods_substitutes/02_gwp_conversion.R`

2.26 4.26 Electrical Equipment

Overview. SF6 is used as an insulating and arc-quenching gas in high-voltage electrical equipment (circuit breakers, switchgear, transformers). Emissions arise from leakage during operation and at end of life. For 1990-1998, emissions are modelled from 1999 base emissions and a RAND global survey. For 1999-2011, emissions are based on Partnership reporting. For 2011-2022, GHGRP Subpart DD data are used. SF6 from military and scientific applications (e.g., AWACS radar) are estimated separately.

2.26.1 Step 1 – Obtain SF6 emissions by source and period

Compile data from the three sources (RAND/modelled pre-1999, Partnership 1999-2011, GHGRP 2011-2022). Military applications use FEMP data (2008+) and IPCC Tier 1 defaults (pre-2008). Scientific applications estimated separately.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	GHGRP Subpart DD; EPA SF6 Partnership; RAND survey; FEMP reports; IPCC Tier 1 defaults for military
Outputs	SF6 emissions (metric tons) by source category and year
R notes	Three-period splice with overlap comparison at transition years (1999, 2011)
Corresponding script	`R/electrical_equipment/01_sf6_emissions.R`

2.26.2 Step 2 – Convert to CO2 equivalents

SF6 x GWP -> MMT CO2 Eq.

[CODING] Coding annotation – Step 2

Item	Detail
Corresponding script	`R/electrical_equipment/02_gwp_conversion.R`

2.27 4.27 SF6 and PFCs from Other Product Uses

Overview. SF6 is used in military applications (AWACS aircraft radar cooling) and scientific research applications (particle accelerators, other lab equipment). PFCs are used in limited other applications. Emissions are estimated using FEMP data and IPCC Tier 1 defaults where FEMP data are unavailable.

2.27.1 Step 1 – Estimate emissions by application

Military SF6: FEMP data (2008+); IPCC Tier 1 default of 740 kg SF6/plane/year for AWACS (pre-2008). Scientific applications estimated from FEMP or IPCC defaults.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	FEMP reports; IPCC Tier 1 AWACS default EF; fleet size data for pre-2008
Outputs	SF6 emissions (metric tons) from military and scientific applications
Corresponding script	`R/other_sf6/01_emissions.R`

2.27.2 Step 2 – Convert to CO2 equivalents

SF6 x GWP -> MMT CO2 Eq.

[CODING] Coding annotation – Step 2

Item	Detail
Corresponding script	`R/other_sf6/02_gwp_conversion.R`

2.28 4.28 Nitrous Oxide from Product Uses

Overview. N2O is used in medical anaesthesia, food processing (aerosol propellant), and other product applications. Upon use, it is released to the atmosphere. Emissions are estimated from national N2O production data, disaggregated by subcategory application and weighted by application-specific emission rates (fraction of N2O consumed that is immediately released).

Emissions = Sum_a (N2O production x share_a x emission rate_a)

2.28.1 Step 1 – Obtain N2O production and disaggregate by application

Annual U.S. N2O production (metric tons) from Census or industry data. Shares by application (medical, food, other) from industry surveys or IPCC defaults.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	N2O production data; application share data
Outputs	N2O production allocated by application
Corresponding script	`R/n2o_product_use/01_production_shares.R`

2.28.2 Step 2 – Apply emission rates and convert to CO2 equivalents

Multiply each application's allocated production x its emission rate -> total N2O emitted. Sum across applications. N2O x GWP -> MMT CO2 Eq.

[CODING] Coding annotation – Step 2

Item	Detail
Inputs	Allocated production by application (Step 1); application-specific emission rates
Outputs	N2O emissions (MMT CO2 Eq.) from product uses
Key operations	Sum of (production x share x emission rate) across applications; GWP conversion
Corresponding script	`R/n2o_product_use/02_emissions_calculation.R`

2.29 4.29 Industrial Processes and Product Use – Summary

Overview. This section aggregates all IPPU subcategory emission estimates into a single chapter total, ensures time-series consistency across sub-categories, and presents the full IPPU emission trend. No new calculations are performed here; this is a consolidation and QC step.

2.29.1 Step 1 – Aggregate all IPPU subcategory estimates

Combine emissions from 4.1-4.28 into a single summary table by gas, subcategory, and year.

[CODING] Coding annotation – Step 1

Item	Detail
Inputs	Outputs from all individual subcategory modules (4.1-4.28)
Outputs	IPPU summary table: MMT CO2 Eq. by subcategory x gas x year
Key operations	`bind_rows()` across subcategory outputs; aggregate to chapter total
R notes	Ideal as a `targets` pipeline terminal target that depends on all upstream subcategory targets
Corresponding script	`R/ippu_summary/01_aggregate.R`

[NOTE] Cross-cutting open questions for IPPU

- **GWP consistency:** All GWP values should be drawn from a single shared lookup table (e.g., `data/gwp_ar5.csv`) used consistently across all subcategories.
- **Energy chapter reconciliation:** Steps that adjust fuel use out of the Energy chapter (coking coal, petroleum coke, natural gas feedstocks for ammonia, etc.) should be reconciled with IPPU module outputs to ensure no double-counting or omission.
- **GHGRP data vintage:** Confirm which reporting year of GHGRP data is used for each subcategory – some subcategories transition from USGS/Census to GHGRP at different points.
- **Missing 2020-2022 data:** Iron and Steel (4.18) and potentially other subcategories use proxy data for recent years. Flag these for review.